

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count      |  |
|-------------------------|--------|------------------------|--|------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 44174      |  |
| Total Threats Mitigated | 16     | Total Packets Analyzed |  | 3991576877 |  |
| High Priority           | 9      | Malicious Packets      |  | 3983726076 |  |
| Low Priority            | 7      | True Negatives Passed  |  | 33154      |  |
| Manual Releases         | 0      |                        |  |            |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count      |  |
|                         |        | ICMP Flood             |  | 0          |  |
| ML Metrics              | Value  | SYN Flood              |  | 0          |  |
| False Positives         | 1      | UDP Flood              |  | 12         |  |
| FP Rate                 | 0.00%  | Uncertain              |  | 4          |  |
| IF Precision            | 99.99% |                        |  |            |  |
| IF Recall               | 95.38% | Action                 |  | Count      |  |
| IF F1-Score             | 97.63% | Quarantined            |  | 3          |  |
| IF Accuracy             | 98.79% | Time Ban               |  | 3          |  |
|                         |        | Blackhole              |  | 0          |  |
|                         |        | Released               |  | 4          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 99.99%  | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 95.38%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 97.63%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 98.79%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 4.62%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 95.38%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 10982          | FN 532            |
| Actual: Normal | FP 1              | TN 32622          |

### 2b. Random Forest: Attack Classification

| Metric    | Value   | Description                                                  |
|-----------|---------|--------------------------------------------------------------|
| Precision | 100.00% | Share of classified flows assigned the correct attack type   |
| Recall    | 100.00% | Share of attacks of each type that were correctly identified |
| F1-Score  | 100.00% | Balanced measure combining Precision and Recall              |
| Accuracy  | 100.00% | Overall proportion of correct classifications                |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 0               | 0              |
| Act: UDP  | 0              | 0               | 10982          |

### 2c. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 243.1 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 3.5 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2d. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 0.97% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 1.98% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 3.37% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.22 | 3        | 3            | 2026-09-15 10:45:57 | 2026-09-15 10:47:19 |
| 10.0.0.20 | 2        | 2            | 2026-09-15 10:45:56 | 2026-09-15 10:47:07 |
| 10.0.0.21 | 2        | 2            | 2026-09-15 10:45:56 | 2026-09-15 10:47:07 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector    | Confidence | Priority | Action        |
|---------------------|-----------|-------|-----------|------------|----------|---------------|
| 2026-09-15 10:34:56 | 10.0.0.20 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:34:56 | 10.0.0.21 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:34:57 | 10.0.0.22 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:35:06 | 10.0.0.20 | DDoS  | UDP Flood | 75.9%      | High     | Time Ban (1m) |
| 2026-09-15 10:35:06 | 10.0.0.21 | DDoS  | UDP Flood | 77.9%      | High     | Time Ban (1m) |
| 2026-09-15 10:35:07 | 10.0.0.20 | DDoS  | UDP Flood | 83.3%      | High     | Time Ban      |
| 2026-09-15 10:35:07 | 10.0.0.21 | DDoS  | UDP Flood | 77.9%      | High     | Time Ban      |
| 2026-09-15 10:35:07 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | High     | Time Ban (1m) |
| 2026-09-15 10:35:08 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | Critical | Time Ban      |
| 2026-09-15 10:36:07 | 10.0.0.20 | DDoS  | UDP Flood | 86.6%      | High     | Released      |
| 2026-09-15 10:36:07 | 10.0.0.21 | DDoS  | UDP Flood | 79.1%      | High     | Released      |
| 2026-09-15 10:36:08 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | High     | Released      |
| 2026-09-15 10:36:18 | 10.0.0.20 | DDoS  | UDP Flood | 86.6%      | Critical | Time Ban (2m) |
| 2026-09-15 10:36:19 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | Critical | Time Ban (2m) |
| 2026-09-15 10:41:51 | 10.0.0.6  | DDoS  | Uncertain | 48.2%      | Low      | Released      |
| 2026-09-15 10:41:59 | 10.0.0.21 | DDoS  | UDP Flood | 83.2%      | High     | Time Ban (2m) |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.22 | UDP Flood | 0.6402   | 85.9% | Critical | Phase 2 | 3        | 11s      | 2026-09-15 10:47:19 | Attack persisted - escalating to time ban |
| 10.0.0.22 | UDP Flood | 0.6132   | 85.9% | High     | Phase 2 | 1        | 1m 11s   | 2026-09-15 10:47:08 | Ban Expired                               |
| 10.0.0.21 | UDP Flood | 0.6399   | 79.0% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-15 10:47:07 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood | 0.6167   | 84.5% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-15 10:47:07 | Attack persisted - escalating to time ban |
| 10.0.0.22 | UDP Flood | 0.6398   | 77.7% | High     | Phase 2 | 1        | 11s      | 2026-09-15 10:46:08 | Attack persisted - escalating to time ban |
| 10.0.0.21 | UDP Flood | 0.6458   | 75.9% | High     | Phase 2 | 1        | 11s      | 2026-09-15 10:46:07 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood | 0.6448   | 84.5% | High     | Phase 2 | 1        | 11s      | 2026-09-15 10:46:07 | Attack persisted - escalating to time ban |

#### Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

\_\_\_\_\_  
SIGNATURE OVER PRINTED NAME  
**Network Administrator**

\_\_\_\_\_  
DATE